第 5 次作业

随机过程

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提交日期	10 月 8 日		

1 Problem 67

At all times, an urn contains N balls—some white balls and some black balls. At each stage, a coin having probability p, $0 < p < 1$, of landing heads is flipped. If heads appears, then a ball is chosen at random from the urn and is replaced by a white ball; if tails appears, then a ball is chosen from the urn and is replaced by a black ball. Let X_n denote the number of white balls in the urn after the nth stage.

1.1

Is $\{X_n, n \geq 0\}$ a Markov chain? If so, explain why.

解：是的，$\{X_n, n \geq 0\}$ 是一个马尔可夫链。因为下一时刻的状态只依赖于当前时刻的状态，而与之前的状态无关。
我们分别计算 $X_n = i, \forall i \in \{0, 1, \ldots, N\}$ 时，下一时刻 X_{n+1} 的状态转移概率：

$$P\{X_{n+1} = i+1 \mid X_n = i\} = p \cdot \frac{N-i}{N}$$

$$P\{X_{n+1} = i-1 \mid X_n = i\} = (1-p) \cdot \frac{i}{N}$$

$$P\{X_{n+1} = i \mid X_n = i\} = p \cdot \frac{i}{N} + (1-p) \cdot \frac{N-i}{N}$$

1.2

What are its classes? What are their periods? Are they transient or recurrent?

解：
该马尔可夫链的状态空间为 $\{0, 1, \ldots, N\}$。所有状态都是互通的，所以链是不可约的，所有状态都属于同一个类。每个状态的周期都是 1，因为存在自循环。由于链是不可

约的且具有有限状态空间，所以所有状态都是常返的。

1.3

Compute the transition probabilities P_{ij}.

解：
由第一问，对于 $i,j \in \{0,1,\ldots,N\}$，转移概率为：

$$P_{ij} = \begin{cases} 1-p+(2p-1)\dfrac{i}{N}, & j=i, \\ p\dfrac{N-i}{N}, & j=i+1,\ i<N, \\ (1-p)\dfrac{i}{N}, & j=i-1,\ i>0, \\ 0, & \text{其他}. \end{cases}$$

1.4

Let $N=2$. Find the proportion of time in each state.

解：
当 $N=2$ 时，状态空间为 $\{0,1,2\}$。转移概率矩阵为：

$$P = \begin{pmatrix} 1-p & p & 0 \\ (1-p)\frac{1}{2} & \frac{1}{2} & p\frac{1}{2} \\ 0 & (1-p) & p \end{pmatrix}$$

寻找长程比例：

$$\pi P = \pi, \quad \sum_{i=0}^{2} \pi_i = 1$$

解得：

$$\pi_0 = (1-p)^2, \quad \pi_1 = 2p(1-p), \quad \pi_2 = p^2$$

1.5

Based on your answer in part (d) and your intuition, guess the answer for the limiting probability in the general case.

解：
我猜测，在一般情况下，极限概率为：

$$\pi_i = \binom{N}{i} p^i (1-p)^{N-i}, \quad i = 0, 1, \ldots, N$$

1.6

Prove your guess in part (e) either by showing that Theorem (4.1) is satisfied or by using the results of Example 4.37.

解：
对于 $N \geq 1$ 马尔可夫链都是不可约的。
令候选分布为 $\pi_k = \binom{N}{k} p^k (1-p)^{N-k}$，$k = 0, 1, \ldots, N$。各分量非负，且由二项式定理，

$$\sum_{k=0}^{N} \pi_k = (p + (1-p))^N = 1.$$

下面验证 $\pi P = \pi$。
当 $k = 0$ 时：

$$\pi_0 P_{00} + \pi_1 P_{10} = \binom{N}{0} p^0 (1-p)^N (1-p) + \binom{N}{1} p^1 (1-p)^{N-1} (1-p) \frac{1}{N} = (1-p)^N = \pi_0$$

当 $k = N$ 时：

$$\pi_{N-1} P_{N-1,N} + \pi_N P_{N,N} = \binom{N}{N-1} p^{N-1} (1-p)^1 p \frac{1}{N} + \binom{N}{N} p^N (p) = p^N = \pi_N$$

当 $1 \leq k \leq N-1$ 时，由于：

$$\begin{aligned}
&\pi_{k-1} P_{k-1,k} \\
=&\binom{N}{k-1} p^k (1-p)^{N-k+1} \cdot \frac{N-k+1}{N} \\
=&\frac{N!}{(k-1)!(N-k+1)!} p^k (1-p)^{N-k+1} \cdot \frac{N-k+1}{N} \\
=&\frac{(N-1)!}{(k-1)!(N-k)!} p^k (1-p)^{N-k+1} \\
=&\binom{N}{k} p^k (1-p)^{N-k} \cdot (1-p) \frac{k}{N} \\
=&\pi_k P_{k,k-1}
\end{aligned}$$

$$
\begin{aligned}
&\pi_{k+1}P_{k+1,k}\\
=&\binom{N}{k+1}p^{k+1}(1-p)^{N-k}\cdot\frac{k+1}{N}\\
=&\frac{N!}{(k+1)!(N-k-1)!}p^{k+1}(1-p)^{N-k}\cdot\frac{k+1}{N}\\
=&\frac{(N-1)!}{(k)!(N-k-1)!}p^{k+1}(1-p)^{N-k}\\
=&\binom{N}{k}p^{k}(1-p)^{N-k}\cdot p\frac{N-k}{N}\\
=&\pi_k P_{k,k+1}
\end{aligned}
$$

所以：

$$\pi_{k-1}P_{k-1,k}+\pi_k P_{k,k}+\pi_{k+1}P_{k+1,k}=\pi_k(P_{k,k-1}+P_{k,k}+P_{k,k+1})=\pi_k$$

综上，$\pi P=\pi$ 的每个分量等式均成立，所以 π 是平稳分布。
接下来证明马尔可夫链是非周期的。对于任意状态 i，有：

$$P_{ii}=p\frac{i}{N}+(1-p)\frac{N-i}{N}>0$$

所以，马尔可夫链是非周期的。马尔可夫链是有限、不可约、非周期的，所以 π 是极限分布。

1.7

If $p=1$, what is the expected time until there are only white balls in the urn if initially there are i white and $N-i$ black?

解：
当 $p=1$ 时，马尔可夫链的转移概率为：

$$P_{i,i+1}=\frac{N-i}{N},\quad P_{i,i}=\frac{i}{N}\quad,\forall i\in\{0,1,\dots N-1\}$$
$$P_{N,N}=1$$

令 T_i 为从状态 i 到状态 N 的期望步数。则有：

$$T_N=0$$
$$T_i=1+\frac{i}{N}T_i+\frac{N-i}{N}T_{i+1},\quad\forall i\in\{0,1,\dots N-1\}$$

于是：

$$T_i=\frac{N}{N-i}+T_{i+1},\quad\forall i\in\{0,1,\dots N-1\}$$

计算得到：

$$T_i = \sum_{j=i}^{N-1} \frac{N}{N-j} = N \sum_{j=1}^{N-i} \frac{1}{j}, \quad \forall i \in \{0, 1, \dots N-1\}$$

Problem 68

(a) Show that the limiting probabilities of the reversed Markov chain are the same as for the forward chain by showing that they satisfy the equations

$$\pi_j = \sum_i \pi_i Q_{ij}$$

解：
在平稳分布的支撑集（即 $\pi_i > 0$ 的状态）上，对于反向马尔可夫链，转移概率为 $Q_{ij} = \frac{\pi_j P_{ji}}{\pi_i}$。我们需要证明 $\pi_j = \sum_i \pi_i Q_{ij}$。

$$\begin{aligned}
\sum_i \pi_i Q_{ij} &= \sum_i \pi_i \frac{\pi_j P_{ji}}{\pi_i} \\
&= \sum_i \pi_j P_{ji} \\
&= \pi_j \sum_i P_{ji} \\
&= \pi_j
\end{aligned}$$

这表明 $\pi_j = \sum_i \pi_i Q_{ij}$，即反向马尔可夫链具有与正向链相同的平稳分布。在题设所涉及的不可约、正常返、非周期条件下，反向链也满足这些条件，故其极限概率与正向链相同。

Problem 70

A total of m white and m black balls are distributed among two urns, with each urn containing m balls. At each stage, one ball is drawn uniformly at random from each urn and the two drawn balls are interchanged. Let X_n denote the number of black balls in urn 1 after the n-th interchange.

1.8

Give the transition probabilities of the Markov chain $\{X_n, n \geq 0\}$.

解：
设 $X_n = k$，则坛 1 中有 k 个黑球、$m-k$ 个白球；坛 2 中有 $m-k$ 个黑球、k 个白球。每步从两坛中各均匀抽一球交换：

$$P_{k,k-1} = \frac{k}{m}\cdot\frac{k}{m} = \left(\frac{k}{m}\right)^2, \quad P_{k,k+1} = \frac{m-k}{m}\cdot\frac{m-k}{m} = \left(\frac{m-k}{m}\right)^2,$$

$$P_{k,k} = \frac{k}{m}\cdot\frac{m-k}{m} + \frac{m-k}{m}\cdot\frac{k}{m} = \frac{2k(m-k)}{m^2}, \quad 0 < k < m,$$

边界 $P_{0,1} = 1$，$P_{m,m-1} = 1$，其余为 0。

1.9

Without computations, what are the limiting probabilities?

解：
稳态时 $2m$ 个球的每种对称安排等可能，坛 1 相当于从 m 黑 m 白中无放回抽取 m 个，故黑球数服从超几何分布，猜测当 $m \geq 2$ 时极限概率为

$$\pi_k = \frac{\binom{m}{k}\binom{m}{m-k}}{\binom{2m}{m}} = \frac{\binom{m}{k}^2}{\binom{2m}{m}}, \quad k = 0, 1, \ldots, m.$$

当 $m = 1$ 时，上式仍给出平稳分布 $(1/2, 1/2)$，但链在两个状态间交替，从固定初态出发不存在极限分布。

1.10

Find the limiting probabilities and show that the chain is time reversible.

解：验证细致平衡方程 $\pi_k P_{k,k+1} = \pi_{k+1} P_{k+1,k}$。由于 $\binom{m}{k+1} = \binom{m}{k}\frac{m-k}{k+1}$，有

$$\binom{m}{k}^2 (m-k)^2 = \binom{m}{k+1}^2 (k+1)^2,$$

两边同除以 $m^2\binom{2m}{m}$ 即得 $\pi_k P_{k,k+1} = \pi_{k+1} P_{k+1,k}$，$k \to k-1$ 同理，$k \to k$ 平凡成立。故链时间可逆，$\pi = \pi P$。
相邻状态间的转移概率均为正，故链不可约。当 $m \geq 2$ 时，存在内部状态 $0 < k < m$，且 $P_{k,k} > 0$（自环），故其周期为 1，由类性质全链非周期。有限不可约非周期链遍历，故上述 π 即为极限概率。
当 $m = 1$ 时，

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

链的周期为 2，从固定初态出发不存在极限分布；但 $\pi = (1/2, 1/2)$ 仍满足细致平衡方程，因此链在平稳分布下仍然时间可逆。